

Tabla de Viabilidad — Upscaling AI en Tiempo Real

580 mediciones · 6 dispositivos · N modelos · N resoluciones · 4 estados de carga
PSNR medido con imagen real del juego (mine.png). Umbral viable ≥30 FPS · Marginal 15–29 · No viable <15.

FSRCNN_x2.pb

Escala ×2 — resolución flexible — backend OpenCV DNN

Dispositivo	Entrada	Salida	FPS idle	Lat ms	PSNR dB	SSIM	Viabilidad	Stress (FPS, Δ%)
CPU_OCV	128×72	256×144	660	1.5	35.5	0.968	✓ Viable	CPU: 536 (-19%) iGPU: 689 (+4%) dGPU: 680 (+3%)
CPU_OCV	256×144	512×288	115	8.7	37.0	0.972	✓ Viable	CPU: 54 (-53%) iGPU: 144 (+25%) dGPU: 128 (+11%)
CPU_OCV	320×180	640×360	56	17.9	36.8	0.972	✓ Viable	CPU: 26 (-54%) iGPU: 67 (+20%) dGPU: 64 (+15%)
CPU_OCV	480×270	960×540	17	60.4	38.1	0.975	~ MARGINAL	CPU: 9 (-45%) iGPU: 19 (+15%) dGPU: 18 (+9%)
CPU_OCV	640×360	1280×720	5	184.5	38.4	0.976	✗ NO	CPU: 4 (-32%) iGPU: 6 (+6%) dGPU: 6 (+3%)
CPU_OCV	640×480	1280×960	4	253.8	—	—	✗ NO	CPU: 3 (-25%) iGPU: 4 (+4%) dGPU: 4 (+1%)
CPU_OCV	800×600	1600×1200	2	413.2	—	—	✗ NO	CPU: 2 (-33%) iGPU: 3 (+5%) dGPU: 2 (+2%)
CPU_OCV	960×540	1920×1080	2	452.5	39.3	0.978	✗ NO	CPU: 2 (-32%) iGPU: 2 (+6%) dGPU: 2 (+4%)
CPU_OCV	1280×720	2560×1440	1	1000.0	40.3	0.980	✗ NO	CPU: 1 (+24%) iGPU: 1 (+3%) dGPU: 1 (-2%)
CPU_OCV	1920×1080	3840×2160	0	2272.7	—	—	✗ NO	CPU: 0 (-7%) iGPU: 0 (+5%) dGPU: 0 (+0%)
iGPU_OCL	128×72	256×144	95	10.5	35.5	0.968	✓ Viable	CPU: 58 (-39%) iGPU: 99 (+4%) dGPU: 86 (-10%)
iGPU_OCL	256×144	512×288	29	34.3	37.0	0.972	~ MARGINAL	CPU: 28 (-4%) iGPU: 32 (+8%) dGPU: 28 (-5%)
iGPU_OCL	320×180	640×360	21	48.4	36.8	0.972	~ MARGINAL	CPU: 15 (-26%) iGPU: 22 (+6%) dGPU: 19 (-10%)
iGPU_OCL	480×270	960×540	10	103.1	38.1	0.975	✗ NO	CPU: 7 (-26%) iGPU: 10 (+5%) dGPU: 9 (-5%)
iGPU_OCL	640×360	1280×720	5	184.5	38.4	0.976	✗ NO	CPU: 5 (-9%) iGPU: 6 (+7%) dGPU: 5 (-10%)
iGPU_OCL	640×480	1280×960	4	247.5	—	—	✗ NO	CPU: 3 (-23%) iGPU: 4 (+9%) dGPU: 4 (-4%)
iGPU_OCL	800×600	1600×1200	3	389.1	—	—	✗ NO	CPU: 2 (-18%) iGPU: 3 (+11%) dGPU: 3 (-2%)
iGPU_OCL	960×540	1920×1080	2	403.2	39.3	0.978	✗ NO	CPU: 2 (-24%) iGPU: 3 (+6%) dGPU: 2 (-4%)
iGPU_OCL	1280×720	2560×1440	1	709.2	40.3	0.980	✗ NO	CPU: 1 (-26%) iGPU: 1 (-4%) dGPU: 1 (-9%)
iGPU_OCL	1920×1080	3840×2160	1	1612.9	—	—	✗ NO	CPU: 0 (-24%) iGPU: 0 (-31%) dGPU: 1 (-6%)
dGPU_OCL	128×72	256×144	138	7.2	35.5	0.968	✓ Viable	CPU: 144 (+4%) iGPU: 91 (-34%) dGPU: 83 (-40%)
dGPU_OCL	256×144	512×288	39	25.3	37.0	0.972	✓ Viable	CPU: 42 (+6%) iGPU: 28 (-30%) dGPU: 36 (-9%)
dGPU_OCL	320×180	640×360	26	39.1	36.8	0.972	~ MARGINAL	CPU: 9 (-65%) iGPU: 14 (-47%) dGPU: 23 (-10%)
dGPU_OCL	480×270	960×540	11	91.2	38.1	0.975	✗ NO	CPU: 12 (+12%) iGPU: 7 (-35%) dGPU: 11 (-4%)
dGPU_OCL	640×360	1280×720	7	150.8	38.4	0.976	✗ NO	CPU: 7 (+2%) iGPU: 6 (-12%) dGPU: 6 (-13%)
dGPU_OCL	640×480	1280×960	5	199.2	—	—	✗ NO	CPU: 5 (+0%) iGPU: 2 (-52%) dGPU: 4 (-15%)
dGPU_OCL	800×600	1600×1200	3	327.9	—	—	✗ NO	CPU: 3 (+5%) iGPU: 1 (-52%) dGPU: 3 (-11%)
dGPU_OCL	960×540	1920×1080	3	357.1	39.3	0.978	✗ NO	CPU: 3 (+8%) iGPU: 1 (-55%) dGPU: 3 (-9%)
dGPU_OCL	1280×720	2560×1440	2	628.9	40.3	0.980	✗ NO	CPU: 2 (+7%) iGPU: 1 (-52%) dGPU: 1 (-13%)
dGPU_OCL	1920×1080	3840×2160	1	1408.5	—	—	✗ NO	CPU: 1 (+4%) iGPU: 0 (-52%) dGPU: 1 (-15%)

FSRCNN_x3.pb

Escala ×3 — resolución flexible — backend OpenCV DNN

Dispositivo	Entrada	Salida	FPS idle	Lat ms	PSNR dB	SSIM	Viabilidad	Stress (FPS, Δ%)
CPU_OCV	128×72	384×216	596	1.7	33.1	0.932	✓ Viable	CPU: 429 (-28%) iGPU: 635 (+6%) dGPU: 558 (-6%)
CPU_OCV	256×144	768×432	115	8.7	34.6	0.945	✓ Viable	CPU: 74 (-36%) iGPU: 152 (+32%) dGPU: 132 (+14%)
CPU_OCV	320×180	960×540	55	18.2	34.9	0.949	✓ Viable	CPU: 29 (-48%) iGPU: 64 (+17%) dGPU: 61 (+11%)
CPU_OCV	480×270	1440×810	16	60.7	36.4	0.955	~ MARGINAL	CPU: 9 (-44%) iGPU: 19 (+12%) dGPU: 15 (-10%)
CPU_OCV	640×360	1920×1080	8	130.2	36.3	0.957	✗ NO	CPU: 4 (-49%) iGPU: 9 (+16%) dGPU: 8 (+5%)
CPU_OCV	640×480	1920×1440	6	179.5	—	—	✗ NO	CPU: 3 (-50%) iGPU: 6 (+14%) dGPU: 6 (+5%)
CPU_OCV	800×600	2400×1800	2	434.8	—	—	✗ NO	CPU: 2 (-28%) iGPU: 2 (+5%) dGPU: 2 (+2%)
CPU_OCV	960×540	2880×1620	2	471.7	37.7	0.964	✗ NO	CPU: 3 (+21%) iGPU: 2 (+5%) dGPU: 2 (+0%)
CPU_OCV	1280×720	3840×2160	1	854.7	38.4	0.968	✗ NO	CPU: 1 (-16%) iGPU: 1 (+4%) dGPU: 1 (-1%)
iGPU_OCL	128×72	384×216	94	10.6	33.1	0.932	✓ Viable	CPU: 62 (-34%) iGPU: 7 (-93%) dGPU: 88 (-6%)

iGPU_OCL	256×144	768×432	29	33.9	34.6	0.945	~ MARGINAL	CPU: 13 (-55%) iGPU: 25 (-16%) dGPU: 26 (-12%)
iGPU_OCL	320×180	960×540	21	48.0	34.9	0.949	~ MARGINAL	CPU: 14 (-35%) iGPU: 9 (-55%) dGPU: 19 (-10%)
iGPU_OCL	480×270	1440×810	10	103.3	36.4	0.955	✗ NO	CPU: 6 (-38%) iGPU: 4 (-54%) dGPU: 9 (-9%)
iGPU_OCL	640×360	1920×1080	5	196.9	36.3	0.957	✗ NO	CPU: 4 (-26%) iGPU: 3 (-43%) dGPU: 5 (-6%)
iGPU_OCL	640×480	1920×1440	4	252.5	—	—	✗ NO	CPU: 4 (-11%) iGPU: 2 (-40%) dGPU: 4 (-8%)
iGPU_OCL	800×600	2400×1800	2	408.2	—	—	✗ NO	CPU: 2 (-23%) iGPU: 2 (-36%) dGPU: 2 (-4%)
iGPU_OCL	960×540	2880×1620	2	416.7	37.7	0.964	✗ NO	CPU: 2 (-28%) iGPU: 2 (-28%) dGPU: 2 (-10%)
iGPU_OCL	1280×720	3840×2160	1	781.2	38.4	0.968	✗ NO	CPU: 1 (+5%) iGPU: 1 (-30%) dGPU: 1 (-4%)
dGPU_OCL	128×72	384×216	141	7.1	33.1	0.932	✓ VIABLE	CPU: 137 (-3%) iGPU: 133 (-6%) dGPU: 83 (-41%)
dGPU_OCL	256×144	768×432	38	26.5	34.6	0.945	✓ VIABLE	CPU: 41 (+8%) iGPU: 18 (-52%) dGPU: 33 (-13%)
dGPU_OCL	320×180	960×540	24	41.6	34.9	0.949	~ MARGINAL	CPU: 26 (+10%) iGPU: 11 (-53%) dGPU: 22 (-10%)
dGPU_OCL	480×270	1440×810	11	87.3	36.4	0.955	✗ NO	CPU: 12 (+2%) iGPU: 5 (-56%) dGPU: 10 (-14%)
dGPU_OCL	640×360	1920×1080	6	162.3	36.3	0.957	✗ NO	CPU: 7 (+9%) iGPU: 3 (-45%) dGPU: 5 (-11%)
dGPU_OCL	640×480	1920×1440	4	233.1	—	—	✗ NO	CPU: 5 (+10%) iGPU: 2 (-48%) dGPU: 4 (-5%)
dGPU_OCL	800×600	2400×1800	3	370.4	—	—	✗ NO	CPU: 3 (+13%) iGPU: 3 (+3%) dGPU: 2 (-8%)
dGPU_OCL	960×540	2880×1620	2	403.2	37.7	0.964	✗ NO	CPU: 3 (+13%) iGPU: 2 (+1%) dGPU: 2 (-6%)
dGPU_OCL	1280×720	3840×2160	1	724.6	38.4	0.968	✗ NO	CPU: 2 (+12%) iGPU: 1 (+5%) dGPU: 1 (-8%)

FSRCNN_x4.pb

Escala ×4 — resolución flexible — backend OpenCV DNN

Dispositivo	Entrada	Salida	FPS idle	Lat ms	PSNR dB	SSIM	Viabilidad	Stress (FPS, Δ%)
CPU_OCV	128×72	512×288	507	2.0	32.4	0.909	✓ VIABLE	CPU: 388 (-23%) iGPU: 530 (+5%) dGPU: 496 (-2%)
CPU_OCV	256×144	1024×576	96	10.4	33.9	0.927	✓ VIABLE	CPU: 78 (-18%) iGPU: 109 (+14%) dGPU: 98 (+2%)
CPU_OCV	320×180	1280×720	19	52.3	34.0	0.931	~ MARGINAL	CPU: 39 (+103%) iGPU: 56 (+194%) dGPU: 54 (+182%)
CPU_OCV	480×270	1920×1080	15	64.7	35.4	0.941	~ MARGINAL	CPU: 11 (-28%) iGPU: 18 (+14%) dGPU: 17 (+8%)
CPU_OCV	640×360	2560×1440	7	148.1	35.5	0.943	✗ NO	CPU: 4 (-45%) iGPU: 8 (+12%) dGPU: 7 (-1%)
CPU_OCV	640×480	2560×1920	5	204.1	—	—	✗ NO	CPU: 3 (-34%) iGPU: 6 (+16%) dGPU: 5 (-7%)
CPU_OCV	800×600	3200×2400	3	334.4	—	—	✗ NO	CPU: 3 (-5%)
CPU_OCV	960×540	3840×2160	3	366.3	36.8	0.953	✗ NO	CPU: 2 (-19%) iGPU: 3 (+16%) dGPU: 3 (+0%)
iGPU_OCL	128×72	512×288	88	11.4	32.4	0.909	✓ VIABLE	CPU: 86 (-2%) iGPU: 21 (-76%) dGPU: 74 (-15%)
iGPU_OCL	256×144	1024×576	27	36.9	33.9	0.927	~ MARGINAL	CPU: 28 (+3%) iGPU: 11 (-61%) dGPU: 25 (-6%)
iGPU_OCL	320×180	1280×720	19	53.2	34.0	0.931	~ MARGINAL	CPU: 20 (+7%) iGPU: 18 (-4%) dGPU: 18 (-3%)
iGPU_OCL	480×270	1920×1080	9	112.9	35.4	0.941	✗ NO	CPU: 9 (+7%) iGPU: 5 (-41%) dGPU: 8 (-8%)
iGPU_OCL	640×360	2560×1440	5	210.5	35.5	0.943	✗ NO	CPU: 5 (+4%) iGPU: 3 (-37%) dGPU: 4 (-7%)
iGPU_OCL	640×480	2560×1920	3	296.7	—	—	✗ NO	CPU: 4 (+9%) iGPU: 3 (-2%) dGPU: 3 (+1%)
iGPU_OCL	800×600	3200×2400	2	448.4	—	—	✗ NO	CPU: 2 (+8%)
iGPU_OCL	960×540	3840×2160	2	471.7	36.8	0.953	✗ NO	CPU: 2 (+5%) iGPU: 2 (-25%) dGPU: 2 (-5%)
dGPU_OCL	128×72	512×288	124	8.1	32.4	0.909	✓ VIABLE	CPU: 119 (-4%) iGPU: 126 (+2%) dGPU: 81 (-35%)
dGPU_OCL	256×144	1024×576	39	25.9	33.9	0.927	✓ VIABLE	CPU: 38 (-3%) iGPU: 37 (-5%) dGPU: 32 (-18%)
dGPU_OCL	320×180	1280×720	25	40.2	34.0	0.931	~ MARGINAL	CPU: 24 (-2%) iGPU: 24 (-5%) dGPU: 21 (-15%)
dGPU_OCL	480×270	1920×1080	11	91.6	35.4	0.941	✗ NO	CPU: 11 (+0%) iGPU: 11 (-3%) dGPU: 9 (-15%)
dGPU_OCL	640×360	2560×1440	5	182.8	35.5	0.943	✗ NO	CPU: 6 (+5%) iGPU: 5 (-2%) dGPU: 5 (-12%)
dGPU_OCL	640×480	2560×1920	4	240.4	—	—	✗ NO	CPU: 4 (+4%) iGPU: 4 (-3%) dGPU: 4 (-10%)
dGPU_OCL	800×600	3200×2400	3	377.4	—	—	✗ NO	CPU: 3 (+4%)
dGPU_OCL	960×540	3840×2160	2	408.2	36.8	0.953	✗ NO	CPU: 3 (+5%) iGPU: 2 (-7%) dGPU: 2 (-18%)

Modelos ONNX/OpenVINO: resolución de entrada fija por arquitectura interna. dGPU_CUDA solo soporta modelos .onnx.

super-resolution-10.onnx

Escala ×3 fijo — entrada 224×224 hardcoded — ONNX/OpenVINO/CUDA

Dispositivo	Entrada	Salida	FPS idle	Lat ms	PSNR dB	SSIM	Viabilidad	Stress (FPS, Δ%)
CPU_OV	224×224	672×672	88	11.4	28.1	0.808	✓ VIABLE	CPU: 56 (-37%) iGPU: 87 (-1%) dGPU: 101 (+15%)

iGPU_OV	224×224	672×672	336	3.0	28.1	0.807	✓ VIABLE	CPU: 300 (-11%) iGPU: 332 (-1%) dGPU: 314 (-6%)
dGPU_CUDA	224×224	672×672	895	1.1	28.1	0.808	✓ VIABLE	CPU: 1016 (+14%) iGPU: 997 (+11%) dGPU: 621 (-31%)

RealESRGAN_x4.onnx

Escala ×3 fijo — entrada 224×224 hardcoded — ONNX/OpenVINO/CUDA

Dispositivo	Entrada	Salida	FPS idle	Lat ms	PSNR dB	SSIM	Viabilidad	Stress (FPS, Δ%)
CPU_OV	224×224	672×672	78	12.8	28.1	0.808	✓ VIABLE	CPU: 57 (-26%) iGPU: 87 (+12%) dGPU: 117 (+50%)
iGPU_OV	224×224	672×672	335	3.0	28.1	0.807	✓ VIABLE	CPU: 304 (-9%) iGPU: 335 (+0%) dGPU: 301 (-10%)
dGPU_CUDA	224×224	672×672	986	1.0	28.1	0.808	✓ VIABLE	CPU: 1032 (+5%) iGPU: 969 (-2%) dGPU: 622 (-37%)

single-image-super-resolution-1032.xml

Escala ×4 fijo — entrada 480×270 hardcoded — solo CPU_OV / iGPU_OV

Dispositivo	Entrada	Salida	FPS idle	Lat ms	PSNR dB	SSIM	Viabilidad	Stress (FPS, Δ%)
CPU_OV	480×270	1920×1080	16	63.9	46.6	0.996	~ MARGINAL	CPU: 11 (-27%) iGPU: 15 (-4%) dGPU: 16 (+3%)
iGPU_OV	320×180	1920×1080	9	116.9	—	—	✗ NO	—
iGPU_OV	480×270	1920×1080	29	34.3	46.6	0.996	~ MARGINAL	CPU: 29 (-1%) iGPU: 28 (-2%) dGPU: 26 (-12%)

FSRCNN OpenVINO IR: mismos pesos que .pb, mismo PSNR, pero compilado con kernels Intel (AVX/VNNI/oneAPI). Eliminada la restricción de resolución fija 224×224.

FSRCNN_x2.xml

Escala ×2 — resolución flexible — FSRCNN convertido a OpenVINO IR (ovc)

Dispositivo	Entrada	Salida	FPS idle	Lat ms	PSNR dB	SSIM	Viabilidad	Stress (FPS, Δ%)
CPU_OV	128×72	144×1	4331	0.2	36.4	0.975	✓ VIABLE	CPU: 693 (-84%) iGPU: 4264 (-2%) dGPU: 2458 (-43%)
CPU_OV	256×144	288×1	1247	0.8	37.8	0.978	✓ VIABLE	CPU: 269 (-78%) iGPU: 1146 (-8%) dGPU: 668 (-46%)
CPU_OV	320×180	360×1	782	1.3	37.5	0.979	✓ VIABLE	CPU: 199 (-75%) iGPU: 409 (-48%) dGPU: 427 (-45%)
CPU_OV	480×270	540×1	254	3.9	38.7	0.980	✓ VIABLE	CPU: 105 (-59%) iGPU: 218 (-14%) dGPU: 158 (-38%)
CPU_OV	640×360	720×1	80	12.5	39.0	0.981	✓ VIABLE	CPU: 47 (-41%) iGPU: 86 (+7%) dGPU: 64 (-20%)
CPU_OV	640×480	960×1	58	17.2	—	—	✓ VIABLE	CPU: 31 (-46%) iGPU: 57 (-1%) dGPU: 42 (-27%)
CPU_OV	800×600	1200×1	32	31.5	—	—	✓ VIABLE	CPU: 20 (-37%) iGPU: 32 (+1%) dGPU: 26 (-18%)
CPU_OV	960×540	1080×1	32	31.6	39.6	0.983	✓ VIABLE	CPU: 20 (-36%) iGPU: 32 (+0%) dGPU: 24 (-23%)
CPU_OV	1280×720	1440×1	17	57.8	40.6	0.984	~ MARGINAL	CPU: 10 (-40%) iGPU: 17 (+0%) dGPU: 14 (-18%)
CPU_OV	1920×1080	2160×1	8	130.4	—	—	✗ NO	CPU: 6 (-24%) iGPU: 8 (+1%) dGPU: 7 (-13%)
iGPU_OV	128×72	144×1	1024	1.0	36.4	0.975	✓ VIABLE	CPU: 1444 (+41%) iGPU: 1555 (+52%) dGPU: 1460 (+43%)
iGPU_OV	256×144	288×1	487	2.1	37.7	0.978	✓ VIABLE	CPU: 468 (-4%) iGPU: 490 (+1%) dGPU: 443 (-9%)
iGPU_OV	320×180	360×1	304	3.3	37.5	0.979	✓ VIABLE	CPU: 287 (-6%) iGPU: 304 (+0%) dGPU: 279 (-8%)
iGPU_OV	480×270	540×1	133	7.5	38.7	0.980	✓ VIABLE	CPU: 129 (-3%) iGPU: 132 (-1%) dGPU: 116 (-13%)
iGPU_OV	640×360	720×1	72	13.9	39.0	0.981	✓ VIABLE	CPU: 71 (-2%) iGPU: 73 (+1%) dGPU: 62 (-14%)
iGPU_OV	640×480	960×1	54	18.5	—	—	✓ VIABLE	CPU: 54 (-1%) iGPU: 55 (+1%) dGPU: 46 (-14%)
iGPU_OV	800×600	1200×1	34	29.1	—	—	✓ VIABLE	CPU: 34 (+0%) iGPU: 35 (+3%) dGPU: 29 (-14%)
iGPU_OV	960×540	1080×1	32	31.5	39.6	0.983	✓ VIABLE	CPU: 31 (-3%) iGPU: 31 (-3%) dGPU: 27 (-14%)
iGPU_OV	1280×720	1440×1	17	58.6	40.5	0.984	~ MARGINAL	CPU: 18 (+3%) iGPU: 17 (+2%) dGPU: 15 (-9%)
iGPU_OV	1920×1080	2160×1	8	124.4	—	—	✗ NO	CPU: 8 (-0%) iGPU: 8 (+0%) dGPU: 7 (-12%)

FSRCNN_x3.xml

Escala ×3 — resolución flexible — FSRCNN convertido a OpenVINO IR (ovc)

Dispositivo	Entrada	Salida	FPS idle	Lat ms	PSNR dB	SSIM	Viabilidad	Stress (FPS, Δ%)
CPU_OV	128×72	216×1	3876	0.3	33.1	0.920	✓ VIABLE	CPU: 642 (-83%) iGPU: 3854 (-1%) dGPU: 2272 (-41%)
CPU_OV	256×144	432×1	1067	0.9	34.5	0.935	✓ VIABLE	CPU: 281 (-74%) iGPU: 1059 (-1%) dGPU: 604 (-43%)
CPU_OV	320×180	540×1	603	1.7	34.7	0.940	✓ VIABLE	CPU: 179 (-70%) iGPU: 355 (-41%) dGPU: 352 (-42%)
CPU_OV	480×270	810×1	198	5.0	36.1	0.948	✓ VIABLE	CPU: 84 (-58%) iGPU: 214 (+8%) dGPU: 144 (-27%)
CPU_OV	640×360	1080×1	78	12.9	36.1	0.949	✓ VIABLE	CPU: 42 (-46%) iGPU: 78 (+0%) dGPU: 53 (-31%)
CPU_OV	640×480	1440×1	55	18.3	—	—	✓ VIABLE	CPU: 31 (-43%) iGPU: 46 (-15%) dGPU: 38 (-31%)
CPU_OV	800×600	1800×1	33	30.7	—	—	✓ VIABLE	CPU: 22 (-31%) iGPU: 30 (-9%) dGPU: 26 (-22%)
CPU_OV	960×540	1620×1	30	33.4	37.3	0.958	~ MARGINAL	CPU: 18 (-42%) iGPU: 29 (-3%) dGPU: 23 (-22%)
CPU_OV	1280×720	2160×1	16	61.7	38.0	0.963	~ MARGINAL	CPU: 10 (-35%) iGPU: 16 (-1%) dGPU: 14 (-16%)

iGPU_OV	128×72	216×1	1166	0.9	33.1	0.920	✓ VIABLE	CPU: 1607 (+38%) iGPU: 1714 (+47%) dGPU: 1521 (+30%)
iGPU_OV	256×144	432×1	456	2.2	34.5	0.935	✓ VIABLE	CPU: 434 (-5%) iGPU: 458 (+0%) dGPU: 403 (-12%)
iGPU_OV	320×180	540×1	287	3.5	34.7	0.940	✓ VIABLE	CPU: 273 (-5%) iGPU: 275 (-4%) dGPU: 258 (-10%)
iGPU_OV	480×270	810×1	125	8.0	36.1	0.947	✓ VIABLE	CPU: 122 (-2%) iGPU: 124 (-1%) dGPU: 102 (-18%)
iGPU_OV	640×360	1080×1	66	15.1	36.0	0.949	✓ VIABLE	CPU: 66 (+0%) iGPU: 66 (+0%) dGPU: 57 (-14%)
iGPU_OV	640×480	1440×1	51	19.7	—	—	✓ VIABLE	CPU: 50 (-1%) iGPU: 44 (-14%) dGPU: 44 (-13%)
iGPU_OV	800×600	1800×1	33	30.3	—	—	✓ VIABLE	CPU: 32 (-2%) iGPU: 32 (-2%) dGPU: 29 (-14%)
iGPU_OV	960×540	1620×1	31	32.6	37.2	0.957	✓ VIABLE	CPU: 29 (-5%) iGPU: 30 (-1%) dGPU: 27 (-13%)
iGPU_OV	1280×720	2160×1	16	61.5	37.9	0.962	~ MARGINAL	CPU: 16 (+2%) iGPU: 16 (-0%) dGPU: 15 (-9%)

FSRCNN_x4.xml

Escala ×4 — resolución flexible — FSRCNN convertido a OpenVINO IR (ovc)

Dispositivo	Entrada	Salida	FPS idle	Lat ms	PSNR dB	SSIM	Viabilidad	Stress (FPS, Δ%)
CPU_OV	128×72	288×1	3523	0.3	32.3	0.893	✓ VIABLE	CPU: 791 (-78%) iGPU: 3508 (-0%) dGPU: 2116 (-40%)
CPU_OV	256×144	576×1	1004	1.0	33.6	0.914	✓ VIABLE	CPU: 272 (-73%) iGPU: 785 (-22%) dGPU: 546 (-46%)
CPU_OV	320×180	720×1	496	2.0	33.8	0.919	✓ VIABLE	CPU: 127 (-74%) iGPU: 444 (-11%) dGPU: 327 (-34%)
CPU_OV	480×270	1080×1	164	6.1	35.1	0.931	✓ VIABLE	CPU: 89 (-46%) iGPU: 169 (+3%) dGPU: 113 (-31%)
CPU_OV	640×360	1440×1	69	14.5	35.2	0.933	✓ VIABLE	CPU: 43 (-37%) iGPU: 69 (-0%) dGPU: 49 (-29%)
CPU_OV	640×480	1920×1	50	19.9	—	—	✓ VIABLE	CPU: 35 (-31%) iGPU: 50 (+0%) dGPU: 36 (-28%)
CPU_OV	960×540	2160×1	28	35.8	36.3	0.945	~ MARGINAL	CPU: 19 (-33%) iGPU: 28 (-0%) dGPU: 21 (-24%)
iGPU_OV	128×72	288×1	1252	0.8	32.3	0.893	✓ VIABLE	CPU: 1509 (+21%) iGPU: 1445 (+15%) dGPU: 1331 (+6%)
iGPU_OV	256×144	576×1	418	2.4	33.6	0.914	✓ VIABLE	CPU: 394 (-6%) iGPU: 418 (+0%) dGPU: 376 (-10%)
iGPU_OV	320×180	720×1	270	3.7	33.8	0.919	✓ VIABLE	CPU: 255 (-6%) iGPU: 264 (-2%) dGPU: 242 (-11%)
iGPU_OV	480×270	1080×1	117	8.6	35.1	0.930	✓ VIABLE	CPU: 112 (-4%) iGPU: 117 (+0%) dGPU: 99 (-15%)
iGPU_OV	640×360	1440×1	60	16.6	35.2	0.933	✓ VIABLE	CPU: 58 (-3%) iGPU: 60 (+0%) dGPU: 52 (-14%)
iGPU_OV	640×480	1920×1	48	21.0	—	—	✓ VIABLE	CPU: 47 (-2%) iGPU: 47 (-1%) dGPU: 41 (-13%)
iGPU_OV	960×540	2160×1	28	35.5	36.4	0.945	~ MARGINAL	CPU: 28 (-1%) iGPU: 28 (+0%) dGPU: 25 (-11%)

Resumen — Combos viables (≥30 FPS) por resolución de salida

Salida	Combos ≥30 FPS	Mejor opción	PSNR (mine.png)
144×1	2	CPU_OV / FSRCNN_x2.xml (4331 FPS)	36.4 dB
216×1	2	CPU_OV / FSRCNN_x3.xml (3876 FPS)	33.1 dB
256×144	3	CPU_OCV / FSRCNN_x2.pb (660 FPS)	35.5 dB
288×1	4	CPU_OV / FSRCNN_x4.xml (3523 FPS)	32.3 dB
360×1	2	CPU_OV / FSRCNN_x2.xml (782 FPS)	37.5 dB
384×216	3	CPU_OCV / FSRCNN_x3.pb (596 FPS)	33.1 dB
432×1	2	CPU_OV / FSRCNN_x3.xml (1067 FPS)	34.5 dB
512×288	5	CPU_OCV / FSRCNN_x4.pb (507 FPS)	32.4 dB
540×1	4	CPU_OV / FSRCNN_x3.xml (603 FPS)	34.7 dB
576×1	2	CPU_OV / FSRCNN_x4.xml (1004 FPS)	33.6 dB
640×360	1	CPU_OCV / FSRCNN_x2.pb (56 FPS)	36.8 dB
672×672	6	dGPU_CUDA / RealESRGAN_x4.onnx (986 FPS)	28.1 dB
720×1	4	CPU_OV / FSRCNN_x4.xml (496 FPS)	33.8 dB
768×432	2	CPU_OCV / FSRCNN_x3.pb (115 FPS)	34.6 dB
810×1	2	CPU_OV / FSRCNN_x3.xml (198 FPS)	36.1 dB
960×540	1	CPU_OCV / FSRCNN_x3.pb (55 FPS)	34.9 dB
960×1	2	CPU_OV / FSRCNN_x2.xml (58 FPS)	—
1024×576	2	CPU_OCV / FSRCNN_x4.pb (96 FPS)	33.9 dB
1080×1	6	CPU_OV / FSRCNN_x4.xml (164 FPS)	35.1 dB
1200×1	2	iGPU_OV / FSRCNN_x2.xml (34 FPS)	—
1440×1	4	CPU_OV / FSRCNN_x4.xml (69 FPS)	35.2 dB
1620×1	1	iGPU_OV / FSRCNN_x3.xml (31 FPS)	37.2 dB
1800×1	2	iGPU_OV / FSRCNN_x3.xml (33 FPS)	—
1920×1	2	CPU_OV / FSRCNN_x4.xml (50 FPS)	—

Stress tags: CPU = stress --cpu 16 · iGPU = bucle OpenVINO GPU · dGPU = bucle ONNX Runtime CUDA
PSNR medido vs upscale bicúbico con mine.png (captura real del juego 1920×1080). Imagen de entrada redimensionada al input_size con INTER_LANCZOS4.